

The Reduced Graph of a Strongly Chordal Graph is Chordal

Maw-Shang Chang*

Ming-Tat Ko†

Abstract

In this note, we define the reduced graph $R(G)$ of a chordal graph G and prove that $R(G)$ is still chordal if G is k -sun-free for $k \geq 4$. As a corollary of the result, the reduced graph of any induced subgraph of a tree power is chordal.

Keywords: Chordal graphs, k -sun-free chordal graphs, Reduced graph.

1 Preliminary

For a graph G , $V(G)$ and $E(G)$ are the vertex and edge sets of G , respectively. A cycle of length k is a sequence of vertices $v_0, v_1, \dots, v_k$, where $v_k = v_0$, and $(v_i, v_{i+1}) \in E(G)$ for $0 \leq i \leq k-1$. A graph G is *chordal* if it contains no induced subgraph which is a cycle of size greater than three. A subset S of $V(G)$ is a *separator* of a connected graph G if $G[V(G) \setminus S]$ has at least two connected components. A separator S of G is *minimal* if any proper subset of S is not a separator of G . A separator S of G is a (u, w) -separator of G if nodes u and w are in different connected components of $G[V(G) \setminus S]$. A (u, w) -separator S is *minimal* if any proper subset of S is not a (u, w) -separator of G . A *minimal vertex separator* is a minimal (u, w) -separator for some vertices u and w . Note that a minimal vertex separator is not necessarily a minimal separator, as a minimal (u, w) -separator may contain a minimal (x, y) -separator for some other vertices x and y .

Let G be a chordal graph. Let $\mathcal{S}_G$ denote the set of minimal vertex separators of G and $\mathcal{K}_G$ the set of the maximal cliques of G . For each node u of G , let $\mathcal{K}_G(u)$ consist of the maximal cliques of G containing u . A *clique tree* of a chordal graph G is a tree $\mathcal{T}_G$ with $V(\mathcal{T}_G) = \mathcal{K}_G$ such that each $\mathcal{K}_G(u)$ with $u \in V(G)$ induces a subtree of $\mathcal{T}_G$.

*Department of Computer Science and Information Engineering, National Chung Cheng University, Ming-Shiun, Chiayi 621, Taiwan, Email: mschang@cs.ccu.edu.tw.

†Institute of Information Science, Academia Sinica, Taipei 115, Taiwan, Email: mtko@iis.sinica.edu.tw.

Lemma 1 ([1–3]) *The following are equivalent.*

1. G is chordal.
2. Every minimal vertex separator of G induces a clique in G .
3. G has a clique tree.

Lemma 2 ([4]) *Let $\mathcal{T}_G$ be a clique tree of a chordal graph G . Then, S is a minimal vertex separator of G if and only if $S = K_1 \cap K_2$ for some edge (K_1, K_2) of $\mathcal{T}_G$.*

The edge (K_1, K_2) is called a *representative edge* of S . Note that a minimal vertex separator may have more than one representative edges. By Lemmas 1 and 2, the intersection of two cliques of a chordal graph is contained in a minimal vertex separator.

A k -sun is a graph formed from a cycle $v_0, \dots, v_{2k}$, by adding edges between vertices of even index so that $\{v_2, v_4, \dots, v_{2k}\}$ induces a clique. A graph is *strongly chordal* if and only if it is chordal and contains no induced k -sun for any $k \geq 3$.

The *reduced graph* $R(G)$ of G is the graph with $V(R(G)) = \cup_{S \in \mathcal{S}_G} S$ and $E(R(G)) = \{(u, v) | u, v \in S \text{ for some } S \in \mathcal{S}_G\}$.

2 Main Theorem

Theorem 1 *If G is a chordal graph has no k -sun for $k \geq 4$ as an induced subgraph then $R(G)$ is a chordal graph.*

Proof. Assume $R(G)$ is not chordal, we want to prove that G is not k -sun-free for $k \geq 4$. Let $\mathcal{T}_G$ be clique tree of G . Let $C = v_0, v_1, \dots, v_k$, $k \geq 4$, be an induced cycle of $R(G)$. Let S_i denote a minimal vertex separator containing v_i and v_{i+1} and $(K_{i,1}, K_{i,2})$ denote a representative edge of S_i for $0 \leq i \leq k-1$. First, we select S_0 and S_1 such that $(K_{0,1}, K_{0,2})$ and $(K_{1,1}, K_{1,2})$ are the closest in $\mathcal{T}_G$. Iteratively, let S_i

be the minimal vertex separator whose representative edge is closest to that of S_{i-1} among those containing v_i and v_{i+1} for $2 \leq i \leq k-1$. Let P_0 be the path in $\mathcal{T}_G$ connecting $(K_{0,1}, K_{0,2})$ and $(K_{1,1}, K_{1,2})$. Without loss of generality, assume that endnodes of P_1 is $K_{0,1}$ and $K_{1,1}$. Iteratively, let P_i be the path connecting $K_{i,1}$ and edge $(K_{i+1,1}, K_{i+1,2})$ and assume that the endnodes of P_i is $K_{i,1}$ and $K_{i+1,1}$ for $1 \leq i \leq k-2$. Let P_{k-1} be the path connecting $K_{k-1,1}$ and $K_{0,1}$. Let $\mathcal{T}'$ be the subtree of $\mathcal{T}_G$ induced by the nodes in P_i , $0 \leq i \leq k-1$. Then the union of P_i , $0 \leq i \leq k-1$, forms a cycle of $\mathcal{T}'$.

First, we prove that $S_i \neq S_j$ for all $i \neq j$. Suppose that $S_i = S_j$ for some $i < j$. Then $v_i, v_{i+1}, v_j, v_{j+1}$ are in S_i . Thus, either (v_i, v_{j+1}) or (v_{i+1}, v_j) is a chord of the induced cycle C , which leads a contradiction.

Second, we prove that P_i and P_{i+1} overlap at most one node, for $0 \leq i \leq k-1$. Suppose that P_i and P_{i+1} overlap more than one node. Without loss of generality, assume that (K_1, K_2) is an edge in $\mathcal{T}_G$ in both P_i and P_{i+1} . So, v_{i+1} and v_{i+2} is in the minimal vertex separator $K_1 \cap K_2$. However, $K_1 \cap K_2$ is closer than S_{i+1} to S_i . It contradicts the assumption on S_i , $1 \leq i \leq k$.

Third, we prove that $\mathcal{T}'$ is a single node. Suppose not. Let (K_1, K_2) be an edge in $\mathcal{T}'$. Since all P_i , $0 \leq i \leq k-1$ form a cycle of $\mathcal{T}'$, there are at least two paths contain the edge (K_1, K_2) , say P_i and P_j , $i < j$. As we have proved, $j > i+1$. Thus, v_{i+1} and v_{j+1} are in the minimal vertex separator $K_1 \cap K_2$. However, (v_{i+1}, v_{j+1}) is a chord of the induced cycle C , which is a contradiction. Hence $\mathcal{T}'$ is a node. That is $K_{i,1}$, $0 \leq i \leq k-1$ are the same maximal clique K . Since $V(C) = \{v_0, v_1, \dots, v_{k-1}\} \subset K$, the k vertices forms a clique K' of G .

Next, we prove that for each i , $0 \leq i \leq k-1$, there exists a vertex w_i in $K_{i,2}$ such that $N_G(w_i) \cap V(C) = \{v_i, v_{i+1}\}$. Since $K_{i,2}$ is a maximal clique of G different from K , there is a vertex $w_i \in K_{i,2}$ not in K . It is obvious that w_i is adjacent to v_i and v_{i+1} . If w_i is adjacent to v_j other than v_i and v_{i+1} , then w_i, v_j, v_i, v_{i+1} are in a maximal clique $\widehat{K} \neq K$ of G . Since $\{v_j, v_i, v_{i+1}\} \subset \widehat{K} \cap K$, they are in a minimal vertex separator of G . Thus, either (v_j, v_i) or (v_j, v_{i+1}) is a chord of the induced cycle C in $R(G)$, which leads a contradiction.

Therefore, we conclude that $\{w_0, \dots, w_{k-1}, v_0, \dots, v_{k-1}\}$ induces a k -sun, $k \geq 4$, in G . $\square$

The reduced graph of a chordal graph G is chordal does not necessarily imply G is a strongly chordal.

Kearney and Corneil prove that a power of a tree is a strongly chordal graph [5]. By the above lemma, we get a corollary that the reduced graph an induced subgraph of a power of a tree is a chordal graph.

References

- [1] P. Buneman, A characterization of rigid circuit graphs, Discrete Math. 9 (1974) 205–212.
- [2] G. A. Dirac, On rigid circuit graphs, Abh. Math. Sem. Univ. Hamburg 25 (1961) 71–76.
- [3] F. Gavril, The intersection graphs of subtrees in trees are exactly chordal graphs, J. Combin. Theory Ser. B 116 (1974) 47–56.
- [4] Chin-Wen Ho and R. C. T. Lee, Counting clique trees and computing perfect elimination schemes in parallel, Infom. Process. Lett. 31 (1989) 61–68.
- [5] P. E. Kearney and D. G. Corneil, Tree powers, J. of Algorithms 29 (1998) 111–131.